



Fig. 2: Centralised vs distributed forwarding in WLAN

at the same time. This makes it likely that during transition, WLAN will need to support both legacy (802.11a/b/g) and 802.11n clients for both 2.4 GHz and 5 GHz.

Plan access-point placements carefully. One of the key benefits of 802.11n is its extended range. However, when doing a site survey, it is important not to rely on this extended range to cut corners and decrease density of access points. Doing so can cause oversubscription and degraded performance for all the connected clients, especially legacy ones.

Generally, it's better to focus on the improved capacity and performance than increased range. Most of the wireless deployments take place with only coverage perspective. For a wireless network it is essential to consider other aspects like capacity planning and context planning.

Provide adequate network infrastructure. Backhaul, power and cabling can lead to more traffic on your wired network, both at the access points and at the core, eventually needing gigabit Ethernet to support traffic loads in excess of 100 Mbps.

Some new 802.11n access points require more than 12.95 watts, which exceeds the limits on regular PoE. Other access points may need two Ethernet cables (and ports) to support them. Well-engineered access points use only a single 802.3af port and require minimal upgrades to the infrastructure. The new 802.3at standard is on the horizon,

but has not been approved yet. Also, cabling should be Cat5e or better, such as Cat6.

Focus on controller-based centralised wireless architectures. Controller-based centralised wireless architectures require all traffic, including security enforcement, to go through the controller. That means many centralised solutions may not be capable of meeting the needs of 802.11n throughputs. The best solution is the one which can support both centralised as well as distributed forwarding based on the traffic conditions.

Centralised forwarding enables all traffic to be routed through the WLAN controller, keeping traffic near central resources. This is best for applications with low bandwidth, for applications that require real-time multi-subnet roaming (such as VoWLAN) and for traffic that requires deep filtering (such as guest access).

Distributed forwarding enables WLAN management traffic to be routed to the WLAN controller but all data traffic is locally bridged at the access point. This is best for networks characterised by high-bandwidth, remote branches. Employee traffic can be locally bridged as their policies are the same whether wireless or wired. Video is another application that should be locally bridged.

Intrusion detection and prevention. Wireless intrusion detection and prevention systems (WIDS/WIPS) require an immediate solution to

detect rogue 802.11n access points. Certain vendors already have equipment that can detect unauthorised 802.11n access points with the current 802.11a/b/g sensors. These solutions detect and locate rogue 802.11n access points. Upgrade to 802.11n sensors only when required.

Select the right clients. A wide variety of 802.11n clients are available today with more on the way as manufacturers standardise chipsets supporting 802.11n. The biggest client considerations are performance and battery life. Although 802.11n consumes more power, the greater bandwidth can mean that client devices must engage the network for shorter periods, limiting the power drain. Finally, purchase only Wi-Fi-certified devices to ensure interoperability and highest performance.

The migration strategy

Over time, 802.11n will become the dominant enterprise LAN technology—not just the dominant wireless LAN technology. From a technical, interoperable and cost-effective perspective, it is ready to go today. However, enterprises interested in 802.11n should think carefully about the long-term strategic implications of their network architecture as well as cost implications that could be borne by something as seemingly innocuous as power consumption.

802.11n is not just an access point replacement decision. Rather, it may have implications throughout the entire wired network. In the worst case scenario, the enterprises may be forced to upgrade their corporate LAN to gigabit Ethernet or their PoE infrastructure to 802.3at to compensate for poorly designed 802.11n products or a poorly executed deployment. Now is the perfect time to plan your migration strategy and evaluate the architectural impact and the available products. ●

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